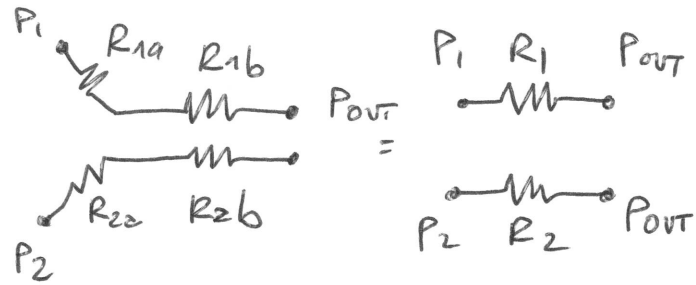
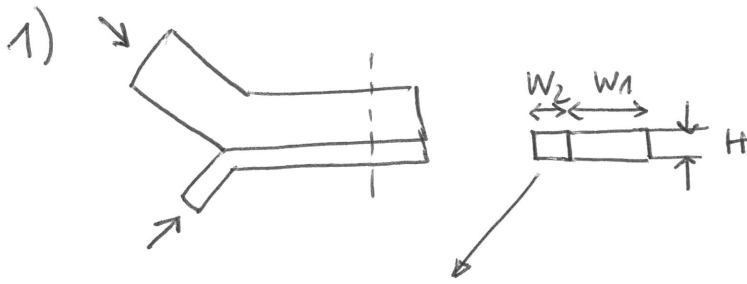


3/7/2020

$$v_{eq} = \frac{2A}{P} = \frac{W_2}{2} = 25 \mu m$$

EXAM



$$R_2 = \frac{8 \cdot \eta \cdot (L_2 + L_3)}{\pi (v_{eq})^4} = \frac{8 \cdot 10^{-3} (12 \mu m)}{\pi (25 \mu m)^4} = \boxed{7.8 \cdot 10^{13}} \left[\frac{Pa \cdot s}{m^3} \right]$$

$$R_1 = \frac{12 \cdot \eta \cdot (L_1 + L_3)}{(1 - 0.63 \frac{H}{W_1}) \cdot H^3 \cdot W_1} = \frac{12 \cdot 10^{-3} (13 \mu m)}{(1 - 0.63 \frac{50}{200}) (50 \mu m)^3 \cdot 200 \mu m} = \boxed{7.4 \cdot 10^{12}} \left[\frac{Pa \cdot s}{m^3} \right]$$

2)

$$V_1 = \frac{Q_1}{W_1 H_1} \quad Q_1 = \frac{\Delta P_1}{R_1} = \frac{0.5 \text{ atm} \cdot 101 \text{ kPa/atm}}{7.4 \cdot 10^{12}}$$

$$V_1 = \frac{6.8 \cdot 10^{-9}}{50 \mu m \cdot 200 \mu m} = 0.68 \frac{m}{s} = \boxed{6.8 \cdot 10^{-9}} \frac{m^3}{s} (6.8 \mu m/s)$$

$$V_2 = \frac{Q_2}{H \cdot W_2} = \frac{\Delta P_2}{R_2} \frac{1}{H W_2} = \frac{10 \text{ kPa}}{7.8 \cdot 10^{13}} \frac{1}{(50 \mu m)^2} = \boxed{0.5 \text{ m/s}}$$

3)

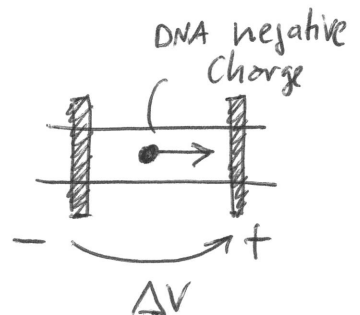
$$F_{drag} = 6 \cdot \pi \cdot r \cdot \eta \cdot v = 1.88 \cdot 10^{-10} [N]$$

4)

$$6 \pi r \cdot \eta \cdot v = Q \cdot E \quad E = \frac{6 \cdot \pi \cdot r \cdot \eta \cdot v}{Q} = 3.5 \cdot 10^5 [V/m]$$

$$Q = \frac{10 \text{ kbp}}{3} \cdot \frac{1.6 \cdot 10^{-19}}{9} = 5.33 \cdot 10^{-16} [C]$$

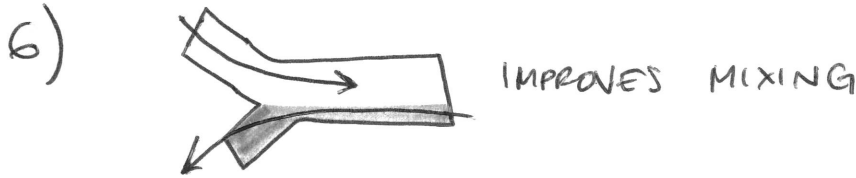
$$\Delta V = E \cdot L_3 = 3.5 \cdot 10^5 \cdot 10 \mu m = \boxed{3534 [V]}$$



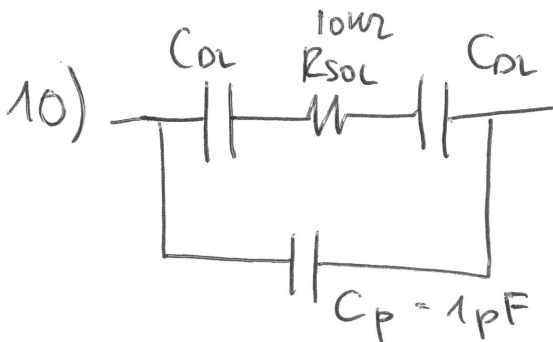
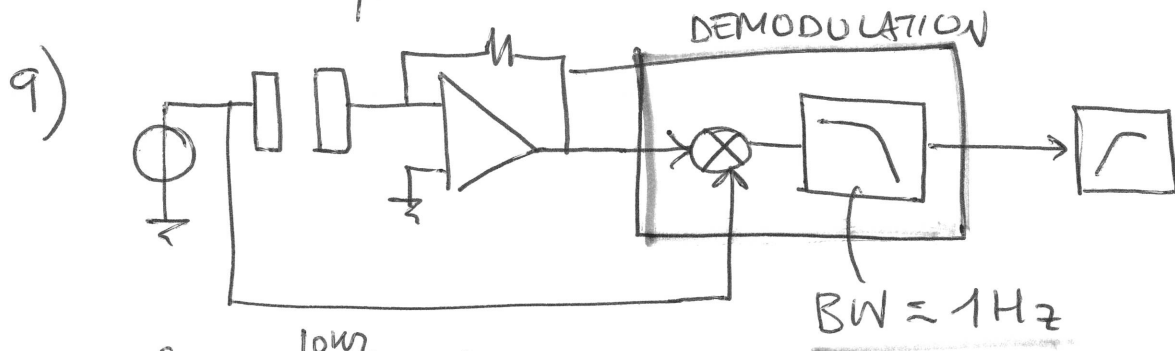
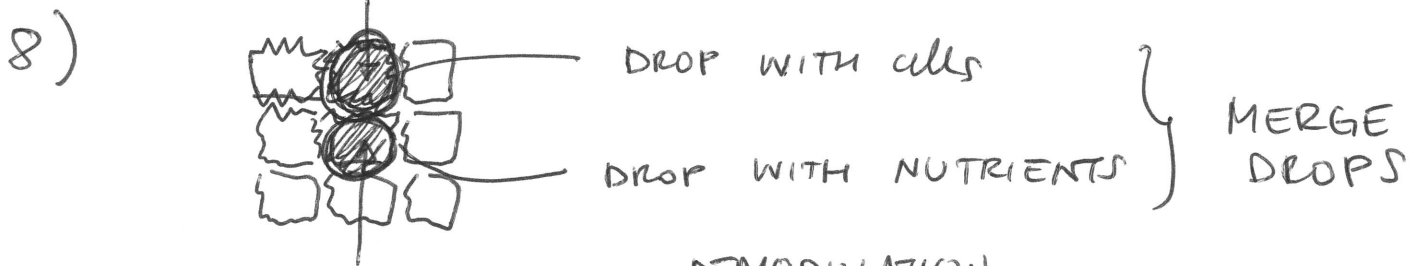
5)

$$D = \frac{kT}{6\pi\eta r} = 4 \cdot 10^{-11} \frac{\text{m}^2}{\text{s}} \rightarrow r = \frac{kT}{6\pi \cdot \eta \cdot 4 \cdot 10^{-11}} = \frac{\frac{26 \text{ mV}}{9}}{6\pi \cdot \eta \cdot 4 \cdot 10^{-11}} \approx 5,5 \cdot 10^{-9}$$

DIAMETER 11 nm



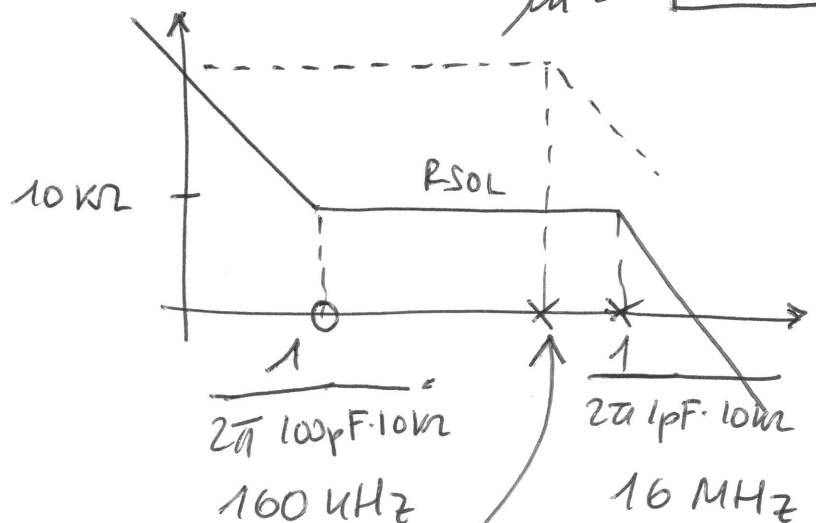
7) PDMS: REPLICA MOLDING
PMMA: EMBOSSING, MOLDING OR MICRO-MILLING (HOT)



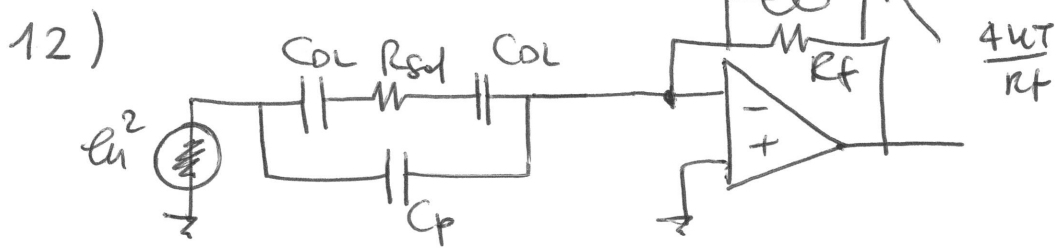
$$C_{0L} = 2000 \mu\text{m}^2 \cdot \frac{0.1 \text{ pF}}{\mu\text{m}^2} = \boxed{200 \text{ pF}}$$

11) $BW|_{\text{total}} = \frac{1}{100 \text{ m}\Omega \cdot 0.2 \text{ pF} \cdot 2\pi}$

$\approx 8 \text{ MHz}$



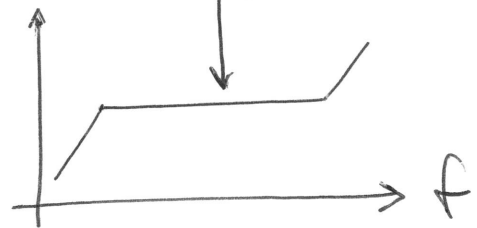
fac between 160 kHz and 8 MHz $\sim$ 1 MHz



$$i_{noise}^2 = \frac{4kT}{R_f}$$

C_{OL} short
 C_p open

white region



$$= \frac{e_{nAC}^2}{R_{sol}^2} = \frac{4kT}{R_f}$$

$$e_{nAC} = R_{sol} \cdot \sqrt{\frac{4kT}{R_f}} = 10k\Omega \cdot \sqrt{\frac{4 \cdot \frac{kT}{9}}{100k\Omega}} = \boxed{4nV/\sqrt{Hz}}$$